

1 Module 1: Course Introduction

1.1 Course Introductory Video

1.1.1 Course Overview

- **Instructors:**
 - **Professor Barsha Mitra** (Co-Instructor)
 - **Professor Lucy J. Gudino** (Co-Instructor)
- **Objective:** Provide a comprehensive understanding of **operating systems (OS)**, their design, functionality, and role in computer systems.

1.1.2 Definition of an Operating System

- **Operating System (OS)** is defined as:
 - A **software** that **manages** and **controls** the **hardware components** of a computer system.
 - Acts as a **resource manager** and **resource allocator**.
 - Provides an **established, convenient, and efficient interface** between:
 - * **User programs** (applications, software)
 - * **Bare hardware** (physical components of the computer)
 - Enables access to a **wide variety of devices**, including:
 - * **Input-Output (I/O) devices** (e.g., printers, digital cameras)
 - * **Multiple processors** (e.g., multi-core CPUs on a single board)

1.1.3 Course Structure

- **Duration:** 10 modules, with one module per week.
- **Learning Progression:**
 - Starts with **foundational concepts** before advancing to **complex topics**.
 - Each module builds on previous knowledge to ensure a **deep understanding** of OS functionality.

1.1.4 Module Breakdown

1.1.4.1 Module 1: Introduction to Computer Components & OS Role

- **Focus:**
 - Understanding the **components of a computer system** (hardware and software).
 - Exploring the **role of the OS** in **controlling and coordinating** these components.
- **Significance:**
 - **Essential foundational knowledge** for grasping **hardware-software interaction**.

1.1.4.2 Module 2: Operating System Components & Services

- **Key Topics:**
 - **Components of an OS** (e.g., kernel, system calls, device drivers).
 - **Services provided by the OS** (e.g., process management, memory management, file systems).
 - **Bootting process:**
 - * What happens when a **computer system boots up**?
 - * How the OS **initializes hardware** and **loads essential software**.

1.1.4.3 Module 3: Process Creation & Management

- **Key Concepts:**

- **Processes** are created when **programs are executed**.
- How the OS **creates, manages, and terminates** processes.
- **Inter-Process Communication (IPC)**:
 - * Mechanisms for **processes to communicate** with each other (e.g., pipes, message queues, shared memory).

1.1.4.4 Module 4: Threads vs. Processes

- **Differentiation:**
 - **Processes vs. Threads:**
 - * **Process**: An **independent program** with its own memory space.
 - * **Thread**: A **lightweight unit of execution** within a process (shares memory with other threads).
- **Thread Management:**
 - How the OS **creates, schedules, and manages** threads.
 - **Advantages of multithreading:**
 - * **Efficiency** (reduced overhead compared to processes).
 - * **Complexity management** in modern computing (e.g., parallel processing).

1.1.4.5 Module 5: Concurrent Process Execution (Multitasking)

- **Focus:**
 - How the OS **executes multiple processes simultaneously**.
 - **Mechanisms enabling multitasking:**
 - * **Time-sharing** (CPU switches between processes rapidly).
 - * **Parallel processing** (multiple CPUs/ cores executing processes at the same time).
- **Importance:**
 - Critical for **handling complex operations** in modern systems.

1.1.4.6 Module 6: Process Scheduling

- **Definition:**
 - **Scheduling** is the method by which the OS **selects processes** for execution and **allocates CPU time**.
- **Key Aspects:**
 - **Scheduling algorithms** (e.g., First-Come-First-Served, Round Robin, Priority Scheduling).
 - **Goals of scheduling:**
 - * **Maximize CPU utilization.**
 - * **Ensure fair resource allocation.**
 - * **Minimize response time** for interactive systems.
- **Relevance:**
 - **Fundamental to multitasking environments** where multiple processes compete for CPU time.

1.1.4.7 Module 7: Deadlocks - Identification & Resolution

- **Definition:**
 - A **deadlock** occurs when **two or more processes** are **blocked forever**, each waiting for a resource held by another.
- **Key Topics:**
 - **Conditions for deadlock** (Coffman Conditions):
 1. **Mutual Exclusion** (only one process can use a resource at a time).
 2. **Hold and Wait** (a process holds a resource while waiting for another).
 3. **No Preemption** (resources cannot be forcibly taken from a process).

- 4. **Circular Wait** (a cycle of processes waiting for each other's resources).
- **Deadlock Prevention Strategies:**
 - * **Avoiding** one of the four conditions.
 - * **Detection & Recovery** (e.g., terminating processes, resource preemption).
- **Importance:**
 - * **Critical for system stability** (prevents indefinite process halts).

1.1.4.8 Module 8: Main Memory Management

- **Focus:**
 - How the OS **divides and allocates main memory (RAM)** to processes.
- **Key Concepts:**
 - **Memory partitioning** (fixed vs. dynamic).
 - **Memory allocation strategies** (e.g., first-fit, best-fit, worst-fit).
 - **Fragmentation** (internal vs. external) and **compaction techniques**.
- **Significance:**
 - **Essential for efficient multitasking** (allows multiple processes to run simultaneously).

1.1.4.9 Module 9: Virtual Memory

- **Definition:**
 - A **memory management technique** that **combines RAM and hard disk space** to create an **illusion of larger memory**.
- **Key Mechanisms:**
 - **Paging** (dividing memory into fixed-size blocks).
 - **Swapping** (moving inactive processes to disk).
 - **Demand Paging** (loading pages only when needed).
- **Advantages:**
 - Enables running **larger and more complex programs** than physical memory alone would allow.
 - Improves **memory efficiency and security**.

1.1.4.10 Module 10: Mass Storage Management

- **Definition:**
 - **Mass storage management** involves the **organization and control of large data volumes** on **secondary storage devices**.
- **Storage Devices Covered:**
 - **Hard Disk Drives (HDDs)**
 - **Solid-State Drives (SSDs)**
 - **Optical Disks (CDs, DVDs)**
 - **Magnetic Tapes**
- **Key Topics:**
 - **File systems** (e.g., FAT, NTFS, ext4).
 - **Disk scheduling algorithms** (e.g., FCFS, SSTF, SCAN).
 - **Data organization & retrieval techniques**.
- **Importance:**
 - **Critical for efficient data storage and access** in modern computing.

1.1.5 Course Pedagogy & Learning Approach

- **Structure:**

- **Easy-paced**, starting with **basic concepts** before progressing to **advanced topics**.
- **Modular design** ensures **logical progression** of knowledge.
- **Encouragement for Learners:**
 - Ideal for students **eager to learn** about OS design and functionality.
 - Instructors express **enthusiasm** and **anticipation** for student engagement.

1.1.6 Closing Remarks

- **Welcome Message:**
 - Instructors **warmly welcome** students and express hope that they will **enjoy the course**.
- **Best Wishes:**
 - Encouragement for **successful learning** and mastery of operating systems.

1.2 Meet Your Instructor - Dr. Lucy J. Gudino

1.2.1 1. Introduction to the Instructor

1.2.1.1 1.1 Personal and Professional Background

- **Name:** Dr. Lucy J. Gudino
- **Current Role:** Professor in the **Department of Computer Science and Information Systems** at **BITS Pilani** (Birla Institute of Technology and Science, Pilani).
- **Academic Journey:**
 - **Bachelor of Engineering (B.E.)** in **Electronics and Communication Engineering** (1994)
 - * Institution: **Kuvempu University**
 - **Master of Technology (M.Tech)** in **Computer Science and Engineering** (2003)
 - * Institution: **Vishveshwaraya Technological University (VTU), Karnataka, India**
 - **Doctor of Philosophy (PhD)** in **Computer Science and Engineering** (2010)
 - * Institution: **Vishveshwaraya Technological University (VTU), Karnataka, India**

1.2.1.2 1.2 Passion for Education

- **Early Inspiration:** Discovered a love for teaching during undergraduate studies while **mentoring peers**.
- **Career Commitment:** Dedicated **29 years** to teaching, with a **lifelong passion for education**.
- **Institutional Affiliation:** Joined **BITS Pilani in 2005**, where she has taught both **graduate and undergraduate students**.

1.2.2 2. Teaching Portfolio

1.2.2.1 2.1 Courses Taught Dr. Gudino has instructed a **diverse range of courses** in **computer communication and systems**, including (but not limited to): - **Introduction to Computing Systems - Distributed Computing - Advanced Computer Organization and Architecture - Cyber-Physical Systems (CPS)** - Additional specialized topics in **computer systems and networking**.

1.2.3 3. Research Expertise

1.2.3.1 3.1 Primary Research Areas Dr. Gudino is an **active researcher** with a focus on advancing technologies in: 1. **Wireless Sensor Networks (WSN)** 2. **Ad Hoc Networks** 3. **Broadband Antenna Design** 4. **Array Signal Processing** 5. **Internet of Things (IoT)**

1.2.3.2 3.2 Doctoral Supervision

- **PhD Mentorship:** Supervised **7 PhD candidates** in research areas including:

- **Wireless Sensor Networks**
- **Antennas for Wireless Applications**

1.2.4 4. Course Introduction: Operating Systems

1.2.4.1 4.1 Role as Instructor

- Dr. Gudino will serve as the **course instructor** for **Operating Systems**.
- **Course Objective:** Provide a **comprehensive understanding** of how **computer system resources** are **managed**.

1.2.4.2 4.2 Significance of the Course

- **Foundational Knowledge:** Serves as a **crucial foundation** for **advanced studies** in:
 - Systems programming
 - Distributed systems
 - Real-time operating systems
 - Embedded systems

1.2.4.3 4.3 Instructor's Commitment to Student Success

- **Engagement:** Encourages **student questions, insights, and perspectives** throughout the semester.
- **Support:** Committed to **fully supporting** the learning experience.
- **Accessibility:** Students are **urged to reach out** with:
 - Academic challenges
 - Questions about course material
 - Any other learning-related concerns
- **Goal:** To **help students succeed** in mastering operating systems concepts.

1.2.4.4 4.4 Closing Remarks

- Expresses **excitement** for the upcoming semester.
- Emphasizes **collaborative learning** and the value of **student contributions**.
- Reinforces **open communication** as a key component of the course.

1.3 Meet Your Instructor - Prof. Barsha Mitra

1.3.1 1. Introduction to the Instructor

1.3.1.1 1.1. Name and Current Position

- **Full Name:** Dr. Barsha Mitra
- **Designation:** Associate Professor
- **Department:** Computer Science and Information Systems
- **Institution:** BITS Pilani, Hyderabad Campus

1.3.1.2 1.2. Educational Background

- **Bachelor's Degree:**
 - **Degree:** Bachelor of Technology (B.Tech)
 - **Field:** Information Technology
 - **University:** West Bengal University of Technology
 - **Year of Completion:** 2009

- **Master's Degree:**
 - **Degree:** Master of Engineering (M.E.)
 - **Field:** Information Technology
 - **University:** Jadavpur University
 - **Year of Completion:** 2011
 - **Additional Note:** Served as a **Teaching Assistant (TA)** during this period, which sparked her passion for teaching.
- **Doctoral Degree:**
 - **Degree:** PhD
 - **Institution:** Indian Institute of Technology (IIT) Kharagpur
 - **Research Domain:** Computer Security
 - **Year of Completion:** 2016

1.3.2 2. Professional Experience

1.3.2.1 2.1. Teaching Career

- **Joined BITS Pilani, Hyderabad Campus:** 2016 (post-PhD)
- **Years of Teaching Experience:** 8 years (as of the lecture)
- **Courses Taught:**
 - Operating Systems
 - Data Structures and Algorithms
 - Object-Oriented Programming
 - Computer Programming
 - Distributed Computing

1.3.2.2 2.2. Mentorship and Student Guidance

- **Undergraduate and Postgraduate Mentorship:**
 - Guided numerous **bachelor's and master's students** in their project work.
- **PhD Supervision:**
 - **Completed Supervision:** 1 PhD student (thesis on **IoT Security**).
 - **Ongoing Supervision:** 3 doctoral students (current).

1.3.3 3. Research Interests

- **Primary Domain:** Computer Security
- **Specific Research Areas:**
 1. **Access Control:** Mechanisms to regulate user permissions and system resource access.
 2. **Malware Detection and Classification:** Techniques to identify and categorize malicious software.
 3. **Intrusion Detection:** Systems to monitor and respond to unauthorized access or attacks.
 4. **IoT Security:** Security challenges and solutions for Internet of Things (IoT) devices and networks.

1.3.4 4. Role in the Operating Systems Course

1.3.4.1 4.1. Course Significance

- **Foundational Course:** Critical for computer science education.
- **Core Objectives:**
 - Understanding **resource management and allocation** in computer systems.
 - Exploring **interactions between hardware and software components**.
 - Providing an **efficient and user-friendly environment** for application execution.

- **Impact on Future Learning:**
 - Serves as a **prerequisite** for advanced topics in computer science.

1.3.4.2 4.2. Instructor's Commitment

- **Teaching Approach:**
 - Aims for an **engaging and enriching learning experience**.
 - Encourages **interactive exploration** of operating system concepts.
- **Student Support:**
 - Will **assist students at every step** of the learning process.
 - Open to **queries and challenges**-students are encouraged to reach out for help.
 - Goal: To **help students succeed** in the course.

1.3.5 5. Closing Remarks

- **Enthusiasm for the Semester:**
 - Expresses eagerness to **interact with students** and delve into operating systems.
- **Call to Action:**
 - Students should **actively participate** and seek guidance when needed.
- **Final Note:**
 - Emphasizes a **collaborative and supportive** learning environment.